

6. Recent trends and Applications.

Q.1 Explain the architecture and training of GAN.

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- A Generative Adversarial Network (GAN), is a deep neural network framework which is able to learn from a set of training data and generate new data with the same characteristics as the training data.
- For example, a GAN trained on photograph of human faces can generate realistic looking faces which are entirely fictitious.
- Architecture :

- A GAN is made up of two neural networks

1) The generator

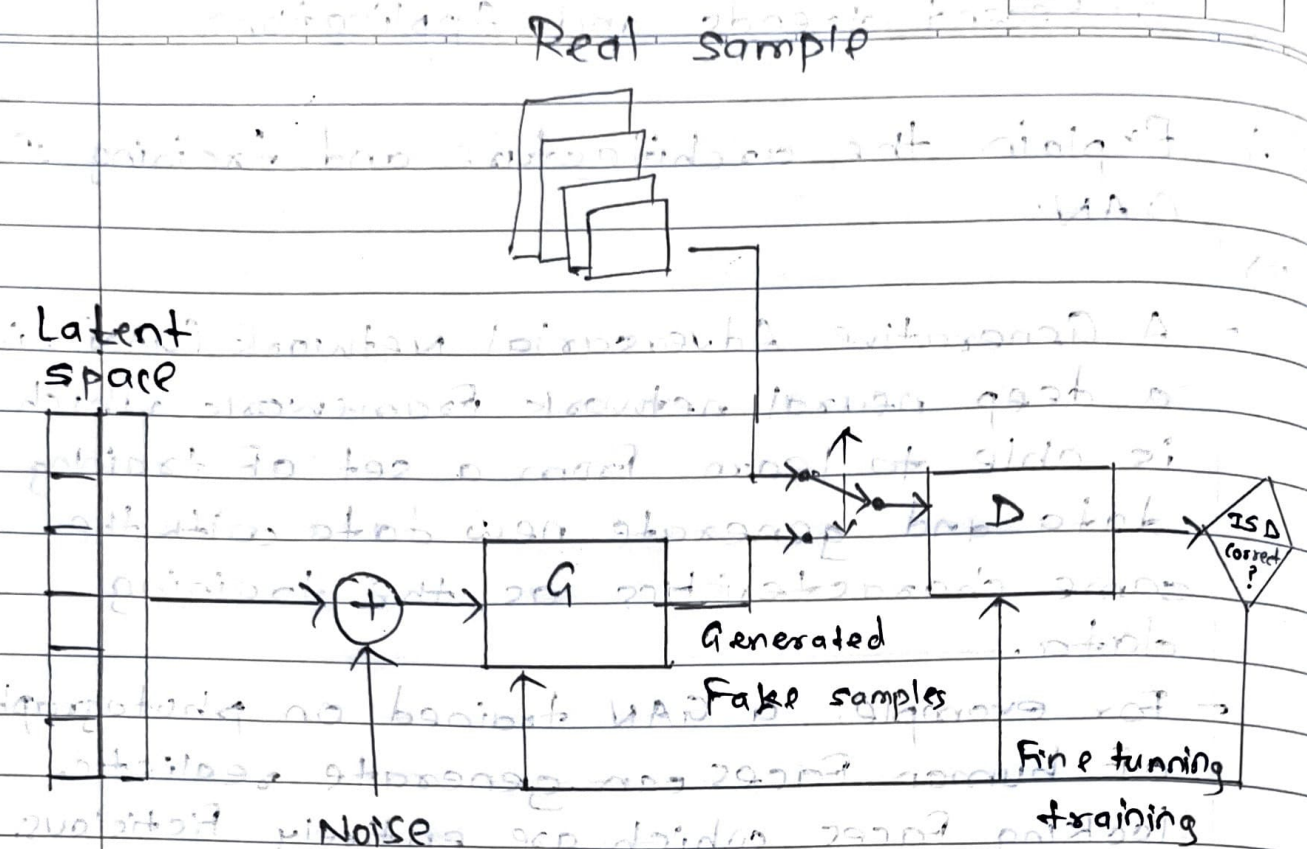
⇒ Which learns to produce realistic fake data from a random seed.

- The fake examples produced by the generator are used as negative examples for training the discriminator.

2) The discriminator

⇒ Which learns to distinguish the fake data from realistic data.

- If the generator produces implausible results, the discriminator penalizes the generator.



- The generator's fake examples, and the training set of examples, are both fed randomly into the discriminator network.
- The discriminator does not know whether a particular input originated from the generator or from the training set.
- To make this Generative and Adversarial process simple, both these blocks are made from DNN based architecture which can be trained through forward and backward propagation technique.

1) Generator's first move

⇒ G takes a random noise vector as input.

- This noise vector contains random values and acts as the starting point for G's creation process.

2) Discriminator Turn

⇒ D's job is to analyze each input and determine whether it's real data or something G cooked up.

- It outputs a probability score between 0 and 1.

1 → likely real

0 → it's Fake

3) Adversarial Learning

⇒ IF the discriminator correctly classifies real data and Fake data as Fake, it strengthens its ability slightly.

- IF the generator successfully fools the discriminator, it receives a positive update while the discriminator is penalized.

4) Generator's Improvement

⇒ Every time the discriminator misclassifies fake data as real, the generator learns and improves.

- Over multiple iterations, the generator produces more convincing synthetic samples.

5) Discriminator's Adoption

⇒ The discriminator continuously refines its ability to distinguish real from Fake data.

6) Training Progression

⇒ As training continues, the generator becomes highly proficient at producing realistic data.

← Eventually, the discriminator struggles to distinguish real from fake, indicating that the GAN has reached a well-trained state.

← At this point, the generator can be used to generate high-quality synthetic data for various applications.

Q.2 Describe the different types of GANs.

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1) Progressive GANs

⇒ In a progressive GAN, the generator's first layer produces very low resolution images, and subsequent layers add detail.

- This technique allows the GAN to train more quickly than comparable non-progressive GANs and produces higher-resolution images.

2) Conditional GANs

⇒ Conditional GANs train on a labeled data set and let you specify the label for each generated instance.

- For example, an unconditional MNIST GAN would produce random digits, while a conditional MNIST GAN would let you specify which digit the GAN should generate.

- Instead of modeling the joint probability $P(X, Y)$, conditional GANs model the conditional probability $P(X|Y)$

3) Image-to-Image translation

⇒ Image-to-Image translation GANs take an image as input and map it to a generated output image with different properties.

- For example, we can take a mask image with blobs of color in the shape of a car, and the GAN can fill in the shape with photorealistic car details

4) Cycle GAN

⇒ Cycle GANs learn to transform images from one set into images that could plausibly belong to another set.

- For example, a Cycle GAN produced the right hand image.

- For e.g., It took an image of a horse and turned it into an image of zebra.

- The training data for the cycle GAN is simply two sets of images.

- The system requires no labels or pairwise correspondences between images.

5) Vanilla GAN

⇒ Vanilla GAN is the simplest type of GAN.

It consists of:

a) The generator and discriminator, both are built using multi-layer perceptrons.

b) The model optimizes its mathematical function formulation using SGD.

Q.3 Illustrate any two applications of GAN.

⇒

- Applications of GANs:

- 1) Image Generation
- 2) Deep Fake
- 3) Image Compression
- 4) Fraud Detection
- 5) Brain-Tumor detection
- 6) Expression identification

1) Deep Fake

⇒ Deepfake is a technique that uses the GAN methods to generate fictitious photographs and videos.

- GANs are a form of deep neural network that has been commonly used to generate deep fake.
- One advantage of GANs that it is capable of learning from a training dataset and creating a sample of data with the same features and characteristics.
- For example, GANs can be used to swipe a "real" image or the video of a person with that of a "fake" one.
- The architecture of GANs consists of two neural network components: generator and discriminator.
- First the model uses the generator to train on a large dataset to create fake data.

- Then, the discriminator is used to learn the fake data from realistic data.
- Many deepfake applications have already been around for quite a few years.
- For e.g., FakeApp, VGGFace.

2) Image Generation

⇒ GANs can create realistic new images from random noise or based on the text prompts, sketches or incomplete images.

- Models like DALL-E generate paintings, sketches and even photorealistic images.
- These models learn patterns from large datasets and generate new images that resemble real-world data.
- Applications include AI-generated artwork, synthetic data creation for training ML models and even designing textures for video games.

3) Fraud Detection

⇒ Financial institutions and cybersecurity firms employ deep learning to detect fraudulent activities in real-time.

- RNNs and LSTM network analyze transaction histories, identifying anomalies that indicate fraud.

- By training on vast datasets of normal and fraudulent transactions, these models can flag suspicious activities with high accuracy.

4) Image Compression

⇒ Deep learning has transformed image compression by improving the efficiency of storing and transmitting visual data while maintaining quality.

- Traditional compression algorithm like JPEG rely on hand-crafted features, whereas AI-driven methods such as Deep Image Prior and Variational Autoencoders learn optimal representations for compact storage.

5) Brain-Tumor Detection

⇒ Deep learning assist in medical diagnostics, particularly in detecting tumors in brain scans with high accuracy.

- AI can detect and classify tumors in early stage.

- Reduces workloads for radiologists by pre-scanning pre-screening cases.